

BANG NA AGROMET, Thailand

WMO: 484530

Lat: 13.6664N

Lon: 100.6061E

Elev: 20

StdP: 14.69

Time Zone: 7.00 (E07)

Period: 08-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	67.9	70.7	54.9	64.2	72.7	58.0	72.0	75.4	10.4	80.8	9.7	81.6	1.9	50	0.365

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
4	12.7	97.5	81.7	96.1	81.4	94.7	80.8	83.7	92.7	83.0	91.9	82.4	91.1	6.6	210

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
81.7	165.0	88.4	80.9	160.4	87.7	80.1	156.4	87.0	47.8	92.8	47.0	91.9	46.2	91.2	86.4

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
10.0	9.4	8.4	DB	63.9	99.2	3.2	2.0	61.6	100.7	59.8	101.9	58.0	103.0	55.7	104.5	(4)
			WB	57.9	84.2	3.9	1.1	55.1	85.0	52.8	85.7	50.6	86.3	47.8	87.1	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	84.8	81.0	83.7	86.0	88.1	87.8	86.4	85.2	85.2	84.7	84.2	83.8	81.4
	DBStd	3.36	3.52	2.62	2.98	2.89	2.84	1.99	1.99	1.72	1.95	2.03	2.37	3.71
	HDD50	0	0	0	0	0	0	0	0	0	0	0	0	0
	HDD65	0	0	0	0	0	0	0	0	0	0	0	0	0
	CDD50	12693	960	943	1115	1143	1170	1091	1092	1092	1040	1059	1014	974
	CDD65	7218	495	523	650	693	705	641	627	627	590	594	564	509
	CDH74	89058	5180	6095	8331	9439	9640	8477	7892	7820	7028	6949	6717	5490
	CDH80	40911	1930	2547	4066	5223	5239	4224	3563	3497	2956	2785	2773	2108
Wind	WSAvg	2.9	2.3	3.2	4.0	3.8	3.3	3.2	3.0	2.9	2.4	1.9	2.4	2.6
Precipitation	PrecAvg	56.8	0.7	0.7	1.8	3.1	7.4	6.7	6.1	7.2	11.5	9.6	1.5	0.4
	PrecMax	73.4	3.0	3.1	6.1	9.3	14.7	13.7	10.0	14.0	19.2	16.5	5.7	2.1
	PrecMin	38.5	0.0	0.0	0.1	0.4	2.1	2.1	1.4	3.2	6.7	1.9	0.0	0.0
	PrecStd	8.8	0.8	0.8	1.7	2.4	3.1	2.9	2.4	3.1	2.9	3.3	1.6	0.5
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	93.0	94.2	96.3	99.0	99.1	97.2	95.2	95.4	94.6	93.8	93.5	93.9
		MCWB	76.3	77.3	78.9	81.3	82.4	81.4	81.1	80.2	80.1	79.8	78.4	77.8
	2%	DB	91.3	92.4	94.6	97.6	97.5	94.7	93.2	93.2	93.1	92.4	92.3	92.0
		MCWB	76.6	77.7	79.7	81.7	82.3	81.4	80.4	80.3	80.2	79.7	78.3	77.4
	5%	DB	89.9	91.1	93.3	96.1	96.0	93.1	91.9	91.5	91.7	91.2	91.2	90.4
		MCWB	75.6	77.8	80.2	81.7	82.0	81.0	80.0	79.9	80.1	79.5	78.0	76.8
	10%	DB	88.3	89.8	91.9	94.6	94.1	91.6	90.6	90.3	90.3	89.9	89.9	88.6
		MCWB	75.1	77.6	79.7	81.3	81.5	80.4	79.6	79.6	79.8	79.2	77.6	75.9
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	80.8	81.4	82.6	85.0	84.9	83.6	82.1	81.9	82.5	82.1	81.8	80.8
		MCDB	88.8	90.0	92.3	94.5	94.7	92.2	91.1	90.7	91.0	89.7	89.3	88.9
	2%	WB	79.5	80.3	81.7	83.8	83.7	82.6	81.5	81.3	81.5	81.2	80.9	79.4
		MCDB	86.5	88.5	91.1	92.8	92.7	91.1	90.0	89.4	89.6	88.3	88.3	87.1
	5%	WB	78.7	79.8	81.1	83.1	83.0	81.8	80.9	80.7	80.9	80.7	80.1	78.6
		MCDB	85.1	87.4	90.0	92.0	91.8	90.0	88.9	88.4	88.6	87.6	87.3	86.0
	10%	WB	77.9	79.2	80.5	82.3	82.3	81.2	80.2	80.1	80.3	80.0	79.3	77.8
		MCDB	84.0	86.2	89.0	91.1	90.8	89.1	87.9	87.4	87.6	86.6	86.0	85.1
Mean Daily Temperature Range	5% DB	MDBR	14.4	13.0	11.9	12.7	12.1	11.0	10.6	11.0	11.5	11.3	11.8	13.4
		MCDBR	15.8	14.2	13.3	13.8	13.9	13.1	12.4	12.6	13.2	12.9	13.1	14.2
		MCWBR	4.7	4.1	3.5	4.1	4.3	4.3	4.2	4.3	4.4	3.9	4.0	4.4
	5% WB	MCDBR	12.5	12.1	12.1	12.5	13.0	12.2	11.4	11.6	12.3	11.9	12.0	13.0
		MCWBR	4.1	3.6	3.6	4.1	4.4	4.5	4.2	4.2	4.6	4.2	4.0	4.1
Clear-Sky Solar Irradiance	taub		0.544	0.610	0.635	0.574	0.498	0.466	0.459	0.460	0.484	0.577	0.524	0.509
	taud		1.963	1.831	1.805	1.992	2.261	2.349	2.369	2.351	2.275	1.972	2.099	2.088
	Ebn at Noon		240	230	228	242	257	262	265	267	263	235	244	246
	Edh at Noon		57	68	71	59	44	40	39	40	44	58	49	49
All-Sky Solar Radiation	RadAvg		1619	1757	1800	1926	1753	1672	1598	1617	1564	1518	1564	1558
	RadStd		93	74	134	121	123	88	82	100	88	84	107	76

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Regional (29 neighbors)	N/A	N/A	N/A	N/A	+0.36	+0.40	N/A	N/A	N/A	N/A

Nomenclature: See separate page